

Introduction to Mathematics and Optimisation

Exercises Week 11

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1 Problem 2

Let A denote the matrix

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ the function given by

$$f(x, y) = x^2 + xy.$$

- (a) Find an invertible matrix B such that

$$B^\top AB = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}.$$

- (b) Explain why A is not positive semidefinite.
(c) Show that f is not a convex function.
(d) Find two vectors $u, v \in \mathbb{R}^2$, such that

$$\begin{aligned} u^\top Au &> 0 \quad \text{and} \\ v^\top Av &< 0 \end{aligned}$$

by possibly using the matrix B from (a).

- (e) Find the critical points for f and account for their types (local minimum/maximum, saddle point etc.).

1.1 Finding Invertible matrix

We can use the method in Section 8.7 to get the invertible matrix B by transforming the extended matrix.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \\ 2 & 0 \\ 0 & 1 \end{pmatrix} \quad (1)$$

So the matrix B is $\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix}$

1.2 Definiteness

By the results of exercise 3.42 diagonal matrix $C = \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$ is positive semi definite if and only if both $x > 0$ and $y > 0$ and by Proposition 8.11 matrix A is positive semidefinite if and only if for any invertible matrix D $D^\top AD$ is positive semidefinite.

And since $B^\top AB$ is not positive semidefinite neither is A

1.3 Convex function

By Theorem 8.23 function f is convex if and only if $\nabla^2 f(x)$ is positive semidefinite for every $x \in \mathbb{R}^2$

The hessian matrix of f is the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ which we previously shown to not be positive semidefinite so f is not a convex function.

1.4 Finding u and v

If we define $y = Bx$ then $y^\top Ay = x^\top B^\top ABx = x^\top (B^\top AB)x = x^\top \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} x$. Then for $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ the equality is positive and for $x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ the equality is negative. Then we get $u = Bx_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v = Bx_2 = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}$

1.5 Critical points

We can use Theorem 6.34 and Theorem 8.12 to find the critical points and determine their types.

```

var('x y')
f = x^2 + x*y
g = f.gradient()
eqs = []
for eq in g:
    eqs.append(eq==0)
solutions = solve(eqs, x, y)

for solution in solutions:
    show(solution)
    h : Matrix = f.hessian().subs(solution)
    if h.is_positive_definite():
        print("Global Minimum")
    elif (-h.subs(solution)).is_positive_definite():
        print("Global Maximum")
    elif not h.is_positive_semidefinite() and not (-h).is_positive_semidefinite():
        print("Saddle Point")
    else:
        print("Nothing")

```

With this we can determine that there is one critical point $[0, 0]$ and it's saddle point